

AMY FAN

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EDUCATION

- University of Arizona** 2023-2025
Master of Science, Statistics and Data Science GPA: 3.78
- Advanced Coursework: Graphs and Networks, Theory of Probability, Statistical Machine Learning, Intro to Deep Learning
 - Academic Awards: Full tuition coverage through competitive teaching opportunities
- University of Arizona** 2019-2023
Bachelor of Science, Statistics and Data Science, Computational Biology GPA: 3.88
- Academic Awards: Magna Cum Laude, Honors, Galileo Circle scholarship \$2500, Data Science Academy Fellow, Wildcat Distinction Tuition Scholarship x4 \$60,000

WORK EXPERIENCE

- Epic Systems**, Server Systems Software Engineer, Operational Data Platform June 2025 - Present
- Developed Go+Gin and Ruby on Rails REST APIs and a React/TypeScript portal for VM fleets, used by 10+ teams.
 - Owned reliability/security foundations: failover design, observability, credentials + OAuth2 pipeline for service backends.
 - Automated authentication-key selection for privileged workflows, cutting average time from 36 minutes to 1 minute per run.
 - Led identification and remediated a configuration vulnerability that enabled OS shell/config overrides; shipped a hardening change enforcing secure defaults for all customers.
 - Provision and troubleshoot customer environments (staging/prod): bootstrap SFTP/SMTP, MFT/NAS integrations, rclone-based transfers, and validate via UAT.
- University of Arizona - Capaldi Lab**, Machine Learning Researcher July 2023 - May 2025
- Architected multi-GPU Mask R-CNN pipelines for noisy microscopy data with custom PyTorch Lightning dataloaders.
 - Demonstrated feasibility of low-light microscopy instance segmentation: achieved 85% mAP/IoU under low-light conditions, within ~5% of reported well-lit SOTA (~87-90%).

LEADERSHIP & ACTIVITIES

- MLH Open Source Hackfest**, Winner October 2024
- Won Best Diversity, Equity & Inclusion at MLH Open Source Hackfest; built *Eyem Watching You*, an eye-tracking mouse controller (OpenCV, dlib, Flask) with multithreaded calibration and smoothing for stable cursor control.

PROJECTS

- Kubernetes Secret Rotation Operator** | Java 21, Java Operator SDK, fabric8, Spring Boot, Helm, Prometheus, GitHub Actions
- Built a Java 21 Kubernetes operator (Java Operator SDK, fabric8) with a level-triggered reconciler managing a custom RotatingSecret CRD; generates credentials via SecureRandom, writes to native Secrets with owner references for cascade deletion, and requeues itself for the next rotation.
 - Instrumented with Micrometer/Prometheus (rotation count + staleness gauge per resource) via Spring Boot Actuator; packaged as a Helm chart; CI publishes image to ghcr.io via GitHub Actions.
- URL Shortener** | Python, FastAPI, Docker, AWS (ECS, ECR, DynamoDB), Terraform
- Built a URL shortener REST API (FastAPI, Pydantic) containerized with Docker, pushed to AWS ECR, and deployed on ECS EC2; selected DynamoDB as backing store after evaluating access patterns (pure key-value, no joins required).
 - Provisioned full AWS infrastructure with Terraform: custom VPC, subnets, Internet Gateway, route tables, security groups, ECS cluster, ECR repository, and DynamoDB table, enabling reproducible one-command deploys and teardowns.
 - Applied least-privilege IAM: separated ECS execution role (ECR pull) from task role scoped to DynamoDB GetItem/PutItem on a single table ARN; validated with a pytest suite covering redirect flow, 404 handling, and Pydantic input validation.
- Eyem Watching You - Eye Tracking to Mousepad** | OpenCV, dlib, Defang, Flask, HTML
- Designed a real-time eye-tracking mouse controller: extracted pupil landmarks (OpenCV/dlib), ran multithreaded calibration + capture, and applied polynomial smoothing for stable cursor control.
 - Developed a Flask web UI and packaged the application for distribution; deployed the web experience using Defang.

SKILLS

Languages: Go, Java, Ruby, Python, TypeScript, JavaScript, SQL, Bash, PowerShell, HTML/CSS, R

Frameworks & Libraries: Gin, Spring Boot, FastAPI, Pydantic, React, Node.js, Flask, PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, dlib, pytest, unittest

Databases & Tools: REST APIs, AWS (ECS, ECR, DynamoDB, VPC, IAM), Terraform, Docker, Helm, Maven, PostgreSQL, MySQL, Kubernetes, GitHub Actions (CI/CD), Prometheus, Linux, Git, Puppet (Hiera), Kerberos, SLURM, Jupyter Notebook, Vercel, Defang, rclone, Vim